Qiyu (Charlie) Yang

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Research Interests

- Representation learning from biological and sensory data: what structure a learned representation of a system should preserve
- Gene regulation as a complex system: how regulatory networks and their dynamics shape cell states & how they change over evolution
- Task-oriented machine learning for clinical reasoning: agents that represent outcome uncertainty faithfully, and prognosis as a trajectory rather than an endpoint

Education

Carnegie Mellon University Pittsburgh, PA
M.S. in Computational Biology *Aug 2025 – May 2027 (expected)*
GPA 3.93/4.33 · Coursework: Machine Learning, Deep Learning, Generative AI, Computational Genomics

Missouri State University & Southwest University Springfield, MO / Chongqing, China
B.S.A. in Plant Science and Technology, 3+1 joint dual-degree program *Sep 2021 – May 2025*
GPA 3.8/4.0 · Minor in Information Security · Dean's List

Publications & Manuscripts

[1] Shi Q†, **Yang Q†**, Huo Y†, Lin Y†, et al. SICH-Copilot: Prior-guided LLM-agent system for intelligent diagnosis and prognostic prediction in spontaneous intracerebral hemorrhage. *npj Digital Medicine* (Collection: Impact of Agentic AI on Care Delivery), revised manuscript under review. (†equal contribution)

[2] Al-Maghrabi L, Lohmann KJ, Yang F, Yang C, **Yang C** (Charlie), Pires A, Ozdoganlar B, Fonseca P, Marques PAAP, LeDuc P. A multi-actuator platform for studying detection of prey-inspired substrate-borne vibrations in semiterrestrial crabs. *Scientific Reports*, revised manuscript under review.

Research Experience

Shenzhen Research Institute of Big Data — AI Large Model Center Shenzhen, China
Research Engineer Intern · LLM-agent system for sICH outcome prediction *May 2026 – Jul 2026*

- Built the agent harness: a uniform tool interface that assembles context from tool schemas and system/user prompts, so new tools plug in unchanged; restructured the ReAct loop as a state machine with evidence gating, tool permissions, and schema-validated fallbacks.
- Made the context state the inherent uncertainty of outcome grading instead of steering the model to an answer.
- Designed SC-RAG, an outcome-aware retrieval module trained with an ordinal triplet loss on clinical–radiomic features; raised retrieval consistency (0.765 → 0.836) and stability (0.650 → 0.791) over text embeddings.
- Evaluated the agent with a 2×2×2 tool ablation (4,800+ reasoning trajectories): the full system beat the best single tool by 7.9% in top-1 balanced accuracy, but tool gains were non-additive and predictions over-tracked the ML prior; traced the remaining error to the limits of admission data, ending further tuning (~$12K saved).
- Drafted the Methods, Results, and Discussion sections of the manuscript and responses to reviewers.

University of Pittsburgh — Miler T. Lee Lab Pittsburgh, PA
Graduate Researcher · Comparative epigenomics of enhancer regulation *Jan 2026 – May 2026*

- Built an end-to-end Slurm-orchestrated pipeline (QC, alignment, peak calling, visualization) for ATAC-seq, ChIP-seq, and WGBS data across multiple mammalian species.
- Defined active enhancers by GMM thresholding of H3K4me1, H3K27ac, and ATAC signal, excluding TSS-proximal regions, and quantified DNA methylation–histone mark associations (Spearman).

Carnegie Mellon University — Philip LeDuc Lab Pittsburgh, PA
Research Assistant · Behavioral time-series analysis of ghost crab locomotion *Sep 2025 – Dec 2025*

- Designed the pause–movement index (PMI) to capture the stop-and-go structure of crab locomotion, an "staircase" curve formed byannotated movement/pause bouts. This metric differed between stimulated and control crabs during and after vibration (GLMM, Holm-corrected).
- Built a Python pipeline that parsed heterogeneous ELAN movement/pause annotation logs into unified time series for per-period.

Southwest University — Kun Lu Lab Chongqing, China
Undergraduate Researcher · BnaA1.CIPK6 regulatory pathways in rapeseed *Mar 2023 – May 2024*

- Processed 60+ RNA-seq samples (QC, alignment, quantification, DESeq2) and ran WGCNA with GO/KEGG enrichment linking CIPK6 expression to JA/SA signaling and carbohydrate synthesis.
- Quantified seed phenotypes with ImageJ and tested morphological differences with ANOVA and multiple-comparison procedures.

PROJECTS

End-to-end speech recognition Transformer
Course project · CMU 11-785 Introduction to Deep Learning

- Implemented and trained a pre-norm encoder–decoder Transformer (8 + 6 layers, RoPE, RMSNorm, SwiGLU, CTC auxiliary loss, ~19M parameters) on LibriSpeech 100h with SpecAugment and mixed precision on PSC GPUs; beam search with a repeat penalty reduced dev CER from 7.45% to 6.75%.

OncoLearn — federated learning for cancer subtyping
CMU × NVIDIA Bio Hackathon

- Built a TCGA-BRCA transcriptome pipeline (991 samples × 19,062 genes), compared engineered features with BulkRNABert embeddings across four classifiers, and co-developed a federated-learning prototype on NVIDIA FLARE.

geneselect — elastic-net gene selection
Course project · CMU Machine Learning

- Implemented sigmoid and softmax elastic-net solvers for subtype gene selection; selected genes beat high-variance and random controls on clustering (ARI/NMI) and overlapped the PAM50 panel; packaged as a pip-installable tool with nested CV, Docker, and CI.

EXPERIENCE

Center for Grapevine Biotechnology, Missouri State University Springfield, MO
Lab Assistant *Sep 2024 – Nov 2024*

- Developed and validated a recombinase polymerase amplification (RPA) assay for *Allorhizobium vitis*, optimizing primers to single-molecule sensitivity, and profiled virulence-plasmid markers across field samples.

Center for Crop Functional Genomics and Molecular Breeding Beijing, China
Germplasm Management and Analysis Assistant *Feb 2024 – Apr 2024*

- Determined transgene copy number in T1 maize by TaqMan qPCR and managed 1,000+ pots of transgenic maize to supply single-copy lines for T2 breeding.

HONORS & AWARDS

Dean's List, Missouri State University	2024
Third-Class Scholarship (top 15%), Southwest University	2023
Second-Class Scholarship (top 9%), Southwest University	2022
Merit Student (3 of 85 in major), Southwest University	2022

LEADERSHIP & SERVICE

Office of Plant Science and Technology, Southwest University Chongqing, China
Student Representative *Mar 2023 – Jul 2024*

- Coordinated teaching and evaluation between Missouri State faculty and students, contributed to English curriculum reform, and interpreted at the SWU–MSU senior seminar.

SKILLS

ML & AI: PyTorch, scikit-learn, AutoGluon, metric/contrastive learning, LLM agents (ReAct, tool use), RAG, ablation and evaluation design
Programming & Computing: Python, R, Go, Bash; Linux, Slurm HPC, Docker, Git, GitHub Actions, Nextflow, Conda
Bioinformatics: RNA-seq (DESeq2), ATAC/ChIP/WGBS, WGCNA, GO/KEGG, FastQC, SAMtools, HISAT2/STAR, BEDtools, deepTools
Languages: Mandarin (native), English (IELTS 8.0)